

Reasoning Behavior Analysis Report

EXPERIMENTAL SETUP

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Data Folder: data/AIME2025__R10/deepseek/p1
Total Items: 300
Total Strategies: 9

OVERALL ACCURACY

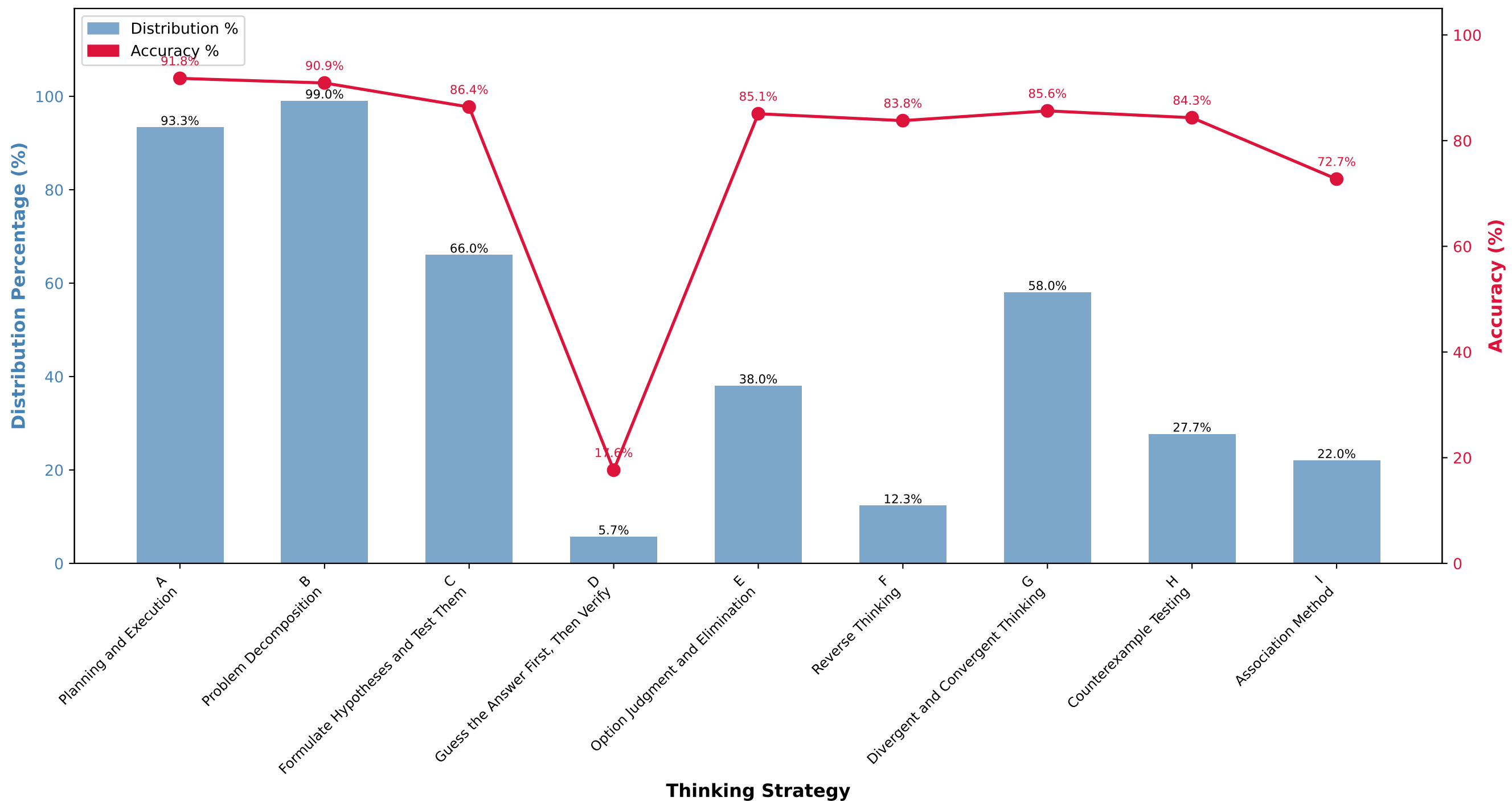
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Total Problems: 300
Correct: 271
Incorrect: 29
Overall Accuracy: 90.33%

THINKING STRATEGY DISTRIBUTION

- =====
- A: Planning and Execution
Distribution: 93.3%
Accuracy: 91.79% (257/280)
 - B: Problem Decomposition
Distribution: 99.0%
Accuracy: 90.91% (270/297)
 - C: Formulate Hypotheses and Test Them
Distribution: 66.0%
Accuracy: 86.36% (171/198)
 - D: Guess the Answer First, Then Verify
Distribution: 5.7%
Accuracy: 17.65% (3/17)
 - E: Option Judgment and Elimination
Distribution: 38.0%
Accuracy: 85.09% (97/114)
 - F: Reverse Thinking
Distribution: 12.3%
Accuracy: 83.78% (31/37)
 - G: Divergent and Convergent Thinking
Distribution: 58.0%
Accuracy: 85.63% (149/174)
 - H: Counterexample Testing
Distribution: 27.7%
Accuracy: 84.34% (70/83)
 - I: Association Method
Distribution: 22.0%
Accuracy: 72.73% (48/66)

Thinking Strategy Distribution and Accuracy



Example for Strategy A: Planning and Execution

PROBLEM

N/A

GROUND TRUTH ANSWER

204

MODEL RESPONSE

Step 1. Understand the Setup

- Alex starts with a disk and draws two perpendicular diameters through the center, dividing the disk into **4 regions**.
- He then draws **25 additional chords**, each by selecting two random points on the perimeter **in different quadrants** and connecting them.
- We aim to compute the **expected number of regions** formed by all 27 segments.

Step 2. Strategy: Adding Chords One by One

When a new chord is added:

- It intersects existing curves at some number m of interior points.
- The chord is divided into $m+1$ segments.
- Each segment splits an existing region into two, adding $m+1$ new regions.

Let R be the total number of regions. Then:

$$R = 4 + \sum_{\text{chords}} (1 + \text{number of interior interse...})$$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Alex divides a disk into four quadrants with two perpendicular diameters intersecting at the center of the disk. He draws 25 more line segments through the disk, drawing each segment by selecting two points at random on the perimeter of the disk in different quadrants and connecting those two points. Find the expected number of regions into which these 27 line segments divide the disk."

Interpretation: Initially, we have a disk. There is a step: "Alex divides a disk into four quadrants with two perpendicular diameters intersecting at the center of the disk." So initially, we have two lines (diameters) that are perpendicular. They divide the disk into 4 regions.

Then: "He draws 25 more line segments through the disk, drawing each segment by selecting two points at random on the perimeter of the disk in different quadrants and connecting those two points." So each additional line segment is drawn. They are chosen as follows: Two points on the perimeter. They are chosen randomly? But careful: "selecting two points at random on the perimeter of the disk in different quadrants" That means: The two endpoints of the line segment are placed on the boundary. And they are chosen such that they lie in different quadrants. So the endpoints are not uniformly distributed along the entire circumference? They are chosen from the perimeter? Possibly: The two points are chosen uniformly and independently along the perimeter? But then condition: They are in different quadrants....

Example for Strategy B: Problem Decomposition

PROBLEM

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N/A

GROUND TRUTH ANSWER

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113

MODEL RESPONSE

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Step 1. Understand the Problem

We have a regular 24-gon with vertex set $\{(S)\}$. We want to choose 12 chords (segments connecting vertices) such that:

- Every vertex is used exactly once.
- All chords have the same length.

Since the polygon is regular, the chord length is determined by the number of steps along the circle between the endpoints. If the vertices are labeled $\{(0, 1, 2, \dots, 23)\}$ (in cyclic order), then a chord connecting vertices $\{(i)\}$ and $\{(j)\}$ has a fixed length if the difference $\{(|i - j| \bmod 24)\}$ is a constant $\{(d)\}$, where $\{(1 \leq d \leq 12)\}$. (Note: $\{(d)\}$ and $\{(24 - d)\}$ give the same chord length.)

Thus, for a fixed $\{(d)\}$, we are looking for a perfect matching of the 24 vertices using only chords of step $\{(d)\}$.

...

CHAIN OF THOUGHT (First 1500 chars)

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We are given: "Let S be the set of vertices of a regular 24-gon. Find the number of ways to draw 12 segments of equal lengths so that each vertex in S is an endpoint of exactly one of the 12 segments."

Interpretation: We have a regular 24-gon. Its vertices are labeled? Possibly they are arranged on a circle. The vertices are points on a circle. We want to choose 12 segments (chords) such that:

- Each vertex is used exactly once.
- All these segments are of equal lengths.
- The segments are drawn. They are chords of the circle. Their lengths: They are chords connecting vertices. They are equal. So if the vertices are numbered $0, \dots, 23$ around the circle, then the chord length is determined by the step. The possible chord lengths? Actually, "equal lengths" means that all these chords are congruent. But note: There are chords that are sides? But then there are chords that are diagonals. But they are all chords. They are equal if the arc between endpoints is the same mod symmetry. Actually, in a regular n -gon, chords of a given step (like from vertex i to $i+k \bmod n$, with $1 \leq k \leq n/2$) are all equal. So here $n=24$, so possible chord lengths correspond to numbers k such that $1 \leq k \leq 12$. But wait: "equal lengths" means that all these 12 segments are of the same chord length. But then the condition "each vertex is an endpoint of exactly one segment" means that we are partitioning the 24 vertices into 12 pairs. And we want the chord (the segment connecting the pair) to have...

Example for Strategy C: Formulate Hypotheses and Test Them

PROBLEM

N/A

GROUND TRUTH ANSWER

104

MODEL RESPONSE

Step 1. Set Up a Coordinate System

Since triangle $\triangle ABC$ is right-angled at A , place:

- $A = (0, 0)$
- $B = (b, 0)$, with $b > 0$
- $C = (0, c)$, with $c > 0$

Then:

$$BC = \sqrt{b^2 + c^2} = 38 \quad \Rightarrow \quad b^2 + c^2 = 1444$$

Step 2. Locate Points K and L

Given:

$$AK = AL = BK = CL = KL = 14$$

Since $AK = AL = 14$, both K and L lie on a circle of radius 14 centered at A . Let:

$$K = (14 \cos \theta, 14 \sin \theta), \quad L = (14 \cos \phi, 14 \sin \phi)$$

with $0 < \theta, \phi < 90^\circ$.

From $BK = 14$:

$$(14 \cos \theta - b)^2 + (14 \sin \theta)^2 = 196$$
$$196 - 28b \cos \theta + b^2 = 196 \quad \Rightarrow \quad \dots$$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Let $\triangle ABC$ be a right triangle with $\angle A = 90^\circ$ and $BC = 38$. There exist points K and L inside the triangle such that $AK = AL = BK = CL = KL = 14$. The area of the quadrilateral $BKLC$ can be expressed as $n\sqrt{3}$ for some positive integer n . Find n ."

Interpretation: We have triangle ABC , right angle at A . $BC = 38$. There exist points K and L inside triangle such that:

$AK = AL = BK = CL = KL = 14$.
So these lengths are all 14. So we have: A, B, C vertices. Points K and L inside triangle. Conditions:
 $AK = 14, AL = 14$, so A is equidistant from K and L . Also $BK = 14$, so B to K is 14. And $CL = 14$, so C to L is 14. And also $KL = 14$.
So essentially, we have two points K and L such that they are at distance 14 from A . And also, K is on circle centered at B with radius 14, and L is on circle centered at C with radius 14. And also distance between K and L is 14.

Also triangle ABC is right at A . And $BC = 38$.

We want area of quadrilateral $BKLC$. That quadrilateral is composed of triangle BKL ? Actually careful: Quadrilateral $BKLC$: vertices B, K, L, C . That is essentially triangle BLC ? Actually it's not necessarily cyclic? Alternatively, it is triangle BLC minus triangle BKL ? Actually, quadrilateral $BKLC$: order: $B \rightarrow K \rightarrow L \rightarrow C$. So it is like two triangles: triangle BKL and triangle BLC ? But careful: Actually, it's a quadrilateral. Its area = $\text{area}(ABC) - \text{area}(\triangle AKL)$ Actually, triangle $A...$

Example for Strategy D: Guess the Answer First, Then Verify

PROBLEM

N/A

GROUND TRUTH ANSWER

240

MODEL RESPONSE

Step 1. Understand the Problem

We consider the function

$$f(x) = \frac{(x-18)(x-72)(x-98)(x-k)}{x}, \quad x > 0,$$

and seek positive values of k such that the **global minimum** of $f(x)$ is attained at **exactly two distinct positive numbers**.

Since $f(x) \rightarrow +\infty$ as $x \rightarrow 0^+$ and $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$, the function must have at least one local minimum. For the global minimum to occur at exactly two points, the function must have **two equal local minima**.

Step 2. Reformulate the Condition

Suppose the minimum value is m , and it is attained at $x = a$ and $x = b$, with $a \neq b$. Then:

$$f(a) = f(b) = m, \quad \text{and} \quad f(x) > m \quad \text{for} \quad x \neq a, b.$$

Define:

$$F(x) = x f(x) - m x = (x-18)(x-72)(x-98)(x-k) - m x.$$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "There are exactly three positive real numbers k such that the function

$$f(x) = \frac{(x-18)(x-72)(x-98)(x-k)}{x}$$

defined over the positive real numbers achieves its minimum value at exactly two positive real numbers x . Find the sum of these three values of k ."

Interpretation: $f(x) = (x-18)(x-72)(x-98)(x-k)/x$, for $x > 0$. We want: There are exactly three positive numbers k such that $f(x)$ achieves its minimum value at exactly two positive real numbers x . That is: For such a k , the function $f(x)$ on $(0, \infty)$ has exactly two distinct points where the minimum is attained. And these are the only k (positive real numbers) that yield that property.

Let's denote: $f(x) = (x-18)(x-72)(x-98)(x-k)/x$. Expand? Alternatively, simplify: $f(x) = (x^4 - (18+72+98+k)x^3 + \dots)/x = x^3 - (18+72+98+k)x^2 + \dots$. Actually careful:

$$f(x) = \frac{(x-18)(x-72)(x-98)(x-k)}{x} = \frac{1}{x} [x^4 - (18+72+98+k)x^3 + (18 \cdot 72 + 18 \cdot 98 + 18 \cdot k + 72 \cdot 98 + 72 \cdot k + 98 \cdot k)x^2 - (18 \cdot 72 \cdot 98 + 18 \cdot 72 \cdot k + 18 \cdot 98 \cdot k + 72 \cdot 98 \cdot k)x + (18 \cdot 72 \cdot 98 \cdot k)].$$

So $f(x) = x^3 - (18+72+98+k)x^2 + (18 \cdot 72 + 18 \cdot 98 + 18 \cdot k + 72 \cdot 98 + 72 \cdot k + 98 \cdot k)x - (18 \cdot 72 \cdot 98 + 18 \cdot 72 \cdot k + 18 \cdot 98 \cdot k + 72 \cdot 98 \cdot k)/x$.

But maybe it's easier to take derivative? Alternatively, consider transformation: $f(x) = (x-18)(x-72)(x-98)(x-k)/x$. For fixed k , we want the function to have exactly two distinct minimizers. That is, the function $f(x)$ on $(0, \infty)$ has a minimum. Possibly it is a polynomial type? Alternatively, maybe we can set $g(x) = f(x) + \text{constant}$? Alternatively, conside...

Example for Strategy E: Option Judgment and Elimination

PROBLEM

N/A

GROUND TRUTH ANSWER

113

MODEL RESPONSE

Step 1. Understand the Problem

We have a regular 24-gon with vertices labeled (in cyclic order) as $\{0, 1, 2, \dots, 23\}$. We want to choose 12 segments (chords) such that:

- Each vertex is used exactly once.
- All segments have the same length.

On a circle, the chord length between two vertices is determined by the smaller arc between them. If the vertices are $\{i\}$ and $\{j\}$, the chord length corresponds to the step:

$$k = \min(|i - j|, 24 - |i - j|), \quad \text{where } 1 \leq k \leq 12$$

Thus, for a fixed $\{k\}$, we want to partition the 24 vertices into 12 pairs such that the vertices in each pair are $\{k\}$ steps apart (or $\{24 - k\}$ steps, which gives the same chord length).

Step 2. Reformulate as a Graph Problem

For a fixed $\{k\}$...

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Let S be the set of vertices of a regular 24-gon. Find the number of ways to draw 12 segments of equal lengths so that each vertex in S is an endpoint of exactly one of the 12 segments."

Interpretation: We have vertices of a regular 24-gon. They are arranged on a circle. We want to choose 12 segments (edges? Actually segments: they are chords? Actually "segments" are drawn between vertices. They are of equal lengths. Which lengths? They are "equal lengths" so all these segments have the same length. But wait: "so that each vertex in S is an endpoint of exactly one of the 12 segments." That means we are partitioning the 24 vertices into 12 pairs. And we are drawing the segment connecting the two vertices in each pair. And we require that all these segments have the same length. And the vertices are on a regular 24-gon. So the possible distances between vertices? On a regular n -gon, the chord lengths are determined by the step number. For a 24-gon, vertices labeled $0, 1, \dots, 23$. The chord between vertices i and j has length corresponding to the minimal step? Actually careful: The vertices are on a circle. The chord length between two vertices separated by k steps (with $1 \leq k \leq 12$, and also symmetric: actually chord length for step k and step $24 - k$ are the same, but since vertices are labeled in order, we usually take k from 1 to 12. But then "equal lengths" means that all chosen pairs correspond to the same chord length. So the problem reduces to: For a given cho...

Example for Strategy F: Reverse Thinking

PROBLEM

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N/A

GROUND TRUTH ANSWER

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821

MODEL RESPONSE

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Step 1. Understand the Problem

We have 12 letters: A, B, C, ..., L (which we can label as numbers 1 through 12). They are randomly grouped into 6 pairs. From each pair, we form a two-letter word by writing the two letters in alphabetical order. Then we list these 6 words in alphabetical order. We want the probability that the **last** (i.e., lexicographically largest) word contains the letter G (which is number 7).

Step 2. Reformulate in Terms of Numbers

Label the letters as:

\\[
A = 1, \\ B = 2, \\ C = 3, \\ D = 4, \\ E = 5, \\ F = 6, \\ G = 7, \\ H = 8, \\ I = 9, \\ J = 10, \\ K = 11, \\ L = 12
\\]

A pair is an unordered set of two numbers. If the pair is $\{a, b\}$ with $a < b$, then the word is ab . The words are listed in lexicographic order (first by the first let...

CHAIN OF THOUGHT (First 1500 chars)

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We are given: "The twelve letters A,B,C,D,E,F,G,H,I,J,K, and L are randomly grouped into six pairs of letters. The two letters in each pair are placed next to each other in alphabetical order to form six two-letter words, and those six words are listed alphabetically. For example, a possible result is AB, CJ, DG, EK, FL, HI. The probability that the last word listed contains G is m/n , where m and n are relatively prime positive integers. Find $m+n$."

Interpretation: We have 12 letters: A, B, C, D, E, F, G, H, I, J, K, L. They are randomly grouped into 6 pairs. So total number of pairings? But careful: The pairs are unordered? Actually, the pairs are sets of two letters. But then from each pair, we form a two-letter word by placing the two letters in alphabetical order. So if the pair is $\{X, Y\}$ with $X < Y$, then the word is XY . And then these six words are listed alphabetically. Alphabetical order: That is the order of the words as strings? They are two-letter words. Their order is determined by the first letter? Actually, careful: The words are like: AB, AC, ... etc. But here the words are exactly pairs from the set. They are listed alphabetically. That means we take the set of these words and order them by the usual lexicographic order. The last word is the one that is greatest in that order. We want the probability that the last word contains the letter G.

Let's denote the pairs as: They are a partition of the set of 12 letters into 6 pairs. The total number of such pairings...

Example for Strategy G: Divergent and Convergent Thinking

PROBLEM

N/A

GROUND TRUTH ANSWER

149

MODEL RESPONSE

Step 1. Find the zeros of $f(x)$

We have

$$f(x) = \sin(7\pi \cdot \sin(5x))$$

Since $\sin(z) = 0$ when $(z = k\pi)$ for any integer (k) , we set:
$$7\pi \cdot \sin(5x) = m\pi \quad \Rightarrow \quad \sin(5x) = \frac{m}{7}$$

where $(m \in \mathbb{Z})$. For real solutions, we require:
$$-1 \leq \frac{m}{7} \leq 1 \quad \Rightarrow \quad -7 \leq m \leq 7$$

So $(m \in \{-7, -6, \dots, 6, 7\})$.

Step 2. Count the number (n) of zeros in $(0, 2\pi)$

Let $(\theta = 5x)$. Then $(x \in (0, 2\pi) \Rightarrow \theta \in (0, 10\pi))$.
We count the number of solutions to $\sin(\theta) = \frac{m}{7}$ in $(0, 10\pi)$.

Case 1: $(m = 0)$

$$\sin(\theta) = 0$$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "There are n values of x in the interval $0 < x < 2\pi$ where $f(x) = \sin(7\pi \cdot \sin(5x)) = 0$. For t of these n values of x , the graph of $y = f(x)$ is tangent to the x -axis. Find $n + t$."

Interpret: $f(x) = \sin(7\pi \cdot \sin(5x)) = 0$.

We want: x in $(0, 2\pi)$ such that $f(x)=0$.

And then among these zeros, some are tangent to x -axis. That means $f(x)=0$ and $f'(x)=0$. And then count: n = number of zeros, t = number of zeros that are tangent (double zeros? Actually careful: "the graph is tangent to the x -axis" means that at that zero, the function touches the axis, so it's a zero of multiplicity > 1 . Typically if $f(x)=0$ and $f'(x)=0$, then it's a point of tangency. But careful: $f(x)=\sin(7\pi \sin(5x))$. So zeros occur when argument is an integer multiple of π ? Actually $\sin(z)=0$ when $z = k\pi$, k integer.

So $f(x)=0 \Leftrightarrow 7\pi \sin(5x) = m\pi$, m integer, so $\sin(5x) = m/7$, and m such that $m/7$ is in $[-1,1]$. So $m \in \{-7, -6, \dots, 6, 7\}$. But careful: m such that $m/7$ is in $[-1,1]$ gives $m = -7, -6, \dots, -1, 0, 1, \dots, 6, 7$. So m from -7 to 7 . But then: $f(x)=0$ if $\sin(5x) = m/7$, m integer.

But then also: $f'(x) = \text{derivative: } f'(x) = \cos(7\pi \sin(5x)) \cdot (7\pi \cos(5x) \cdot 5) = 35\pi \cos(5x) \cos(7\pi \sin(5x))$.

At zeros, $f(x)=0$ so $7\pi \sin(5x) = m\pi$, so $\sin(5x) = m/7$. Then $f'(x) = 35\pi \cos(5x) \cos(m\pi) = 35\pi \cos(5x) \cdot (-1)^m$. Actually $\cos(m\pi) = (-1)^m$.

So $f'(x) = 35\pi \cos(5x) \cdot (-1)^m$.

For tangency, we need $f'(x)=0$. So then we require $\cos(5x)=0$. But also $f(x)=0$. So conditions: ...

Example for Strategy H: Counterexample Testing

PROBLEM

N/A

GROUND TRUTH ANSWER

510

MODEL RESPONSE

Step 1. Rewrite the Inequalities

We consider points $((x, y, z))$ in space that lie on the plane

$$\begin{aligned} & x + y + z = 75 \\ & \text{and satisfy the inequalities} \end{aligned}$$

$$\begin{aligned} & x - yz < y - zx < z - xy. \end{aligned}$$

This gives two strict inequalities:

1. $x - yz < y - zx$
2. $y - zx < z - xy$

Step 2. Simplify the Inequalities

First inequality:

$$\begin{aligned} & x - yz < y - zx \\ & \Rightarrow x - yz - y + zx < 0 \\ & \Rightarrow x(1 + z) - y(1 + z) < 0 \\ & \Rightarrow (1 + z)(x - y) < 0 \end{aligned}$$

Second inequality:

$$\begin{aligned} & y - zx < z - xy \\ & \Rightarrow y - zx - z + xy < 0 \\ & \Rightarrow y(1 + x) - z(1 + x) < 0 \\ & \Rightarrow (1 + x)(y - z) < 0 \end{aligned}$$

Thus, the conditions become:

- $(1 + z)(x - y) < 0$
- $(1 + x)(y - z) < 0$

Step 3. Change of Variables

Let
$$A = 1 + x, \quad B = 1 + y, \dots$$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "The set of points in 3-dimensional coordinate space that lie in the plane $x+y+z=75$ whose coordinates satisfy the inequalities $x-yz < y-zx < z-xy$ forms three disjoint convex regions. Exactly one of those regions has finite area. The area of this finite region can be expressed in the form $a\sqrt{b}$, where a and b are positive integers and b is not divisible by the square of any prime. Find $a+b$."

Interpretation: Points in $\mathbb{R}^3$ that lie on plane: $x+y+z = 75$. And they satisfy:

- (1) $x - yz < y - zx$
 - (2) $y - zx < z - xy$
- So inequalities:
 $x - yz < y - zx \Leftrightarrow x - yz - y + zx < 0 \Leftrightarrow x(1+z) - y(1+z) < 0$? Actually careful:

$$\begin{aligned} & x - yz < y - zx \\ \Rightarrow & x - yz - y + zx < 0 \\ \Rightarrow & x + zx - y - yz < 0 \\ \Rightarrow & x(1+z) - y(1+z) < 0 \\ \Rightarrow & (1+z)(x-y) < 0. \end{aligned}$$

So inequality (1) becomes: $(1+z)(x-y) < 0$.

Similarly, second: $y - zx < z - xy$

$$\begin{aligned} \Rightarrow & y - zx - z + xy < 0 \\ \Rightarrow & y + xy - z - zx < 0 \\ \Rightarrow & y(1+x) - z(1+x) < 0 \\ \Rightarrow & (1+x)(y-z) < 0. \end{aligned}$$

And third? Actually there are only two inequalities given: $x-yz < y-zx$ and $y-zx < z-xy$. But then there is also implicitly? Actually the ordering: $x-yz < y-zx < z-xy$. That gives two inequalities. So we have:

- (1) $(1+z)(x-y) < 0$
- (2) $(1+x)(y-z) < 0$

Wait check: For the second: $y - zx < z - xy \Rightarrow y - zx - z + xy < 0 \Rightarrow y(1+x) - z(1+x) = (1+x)(y-z) < 0$. Yes.

So conditions:

- $(1+z)(x-y) < 0$,
- $(1+x)(y-z) < 0$.

Also note: There is no condition from the third pair? Actually the full ordering is: $x-yz < y-zx$ and $y-zx < z-xy$. That gives these two. But then there is also...

Example for Strategy I: Association Method

PROBLEM

N/A

GROUND TRUTH ANSWER

468

MODEL RESPONSE

Step 1. Set Up Coordinates on the Line

Since the points lie in order along a straight line, assign coordinates as follows:

- Let $A = 0$
- Let $B = b$
- Let $C = c$
- Let $D = d$
- Let $E = e$
- Let $F = f$

Given:

- $AC = 26 \Rightarrow c = 26$
- $AF = 73 \Rightarrow f = 73$
- $CE = 31 \Rightarrow e = c + 31 = 57$
- $DF = 33 \Rightarrow d = f - 33 = 40$
- $BD = 22 \Rightarrow b = d - 22 = 18$

Coordinates:

- $A = 0$
- $B = 18$
- $C = 26$
- $D = 40$
- $E = 57$
- $F = 73$

Step 2. Locate Point G

Let $G = (x_G, y_G)$, with $y_G \neq 0$. Given:

- $CG = 40$
- $DG = 30$

Using distance formula:

$$(x_G - 26)^2 + y_G^2 = 40^2$$
$$(x_G - 40)^2 + y_G^2 = 30^2$$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Six points A, B, C, D, E, F lie in a straight line in that order. Suppose that G is a point not on the line and that $AC = 26$, $BD = 22$, $CE = 31$, $DF = 33$, $AF = 73$, $CG = 40$, and $DG = 30$. Find the area of triangle BGE."

Let coordinates: Place line along x-axis. Let coordinates: A, B, C, D, E, F. Order: A, B, C, D, E, F. Given distances: $AC = 26$, so if $A=0$, then $C=26$. But careful: order: A, B, C, D, E, F. So positions: A, B, C, D, E, F. Let $A=0$, $B = b$, $C = c$, $D = d$, $E = e$, $F = f$. Then given: $AC = c - 0 = 26$ so $c=26$.

$BD = d - b = 22$.

$CE = e - c = 31$ so $e = 26+31 = 57$.

$DF = f - d = 33$.

$AF = f - 0 = 73$ so $f=73$.

Then from DF: $f-d = 33$ so $d = f - 33 = 73-33 = 40$.

Then $BD = d - b = 22$ so $b = d - 22 = 40-22 = 18$.

So coordinates: $A=0$, $B=18$, $C=26$, $D=40$, $E=57$, $F=73$.

Now point G is not on line. Given: $CG = 40$, $DG = 30$. So coordinates of G? Let $G = (x_G, y_G)$ with $y_G \neq 0$. We know distances from C and D. C at $(26,0)$, D at $(40,0)$. So:

$CG^2 = (x_G - 26)^2 + y_G^2 = 40^2 = 1600$.

$DG^2 = (x_G - 40)^2 + y_G^2 = 30^2 = 900$.

Subtract: $(x_G - 26)^2 - (x_G - 40)^2 = 1600 - 900 = 700$.

Compute: $(x^2 - 52x + 676) - (x^2 - 80x + 1600) = 28x - 924 = 700$, so $28x = 1624$, $x = 1624/28 = 58$? Actually $28*58 = 1624$, so $x_G = 58$.

Then plug into DG^2 : $(58-40)^2 + y_G^2 = 18^2 + y_G^2 = 324 + y_G^2 = 900$, so $y_G^2 = 576$, so $y_G = \pm 24$. Choose positive: $y_G = 24$.

So $G = (58, 24)$.

We want area of triangle BGE. $B = (18,0)$, $G = (58,24)$, $E = (57,0)$. So triangle vertices: $(18,0)$, $(58,24)$, $(57,0)$. Compute area. Coordina...